

Demonstrating Flower Jelly Printer for Parametrically Designed Flower Jelly

Mako Miyatake
The University of Tokyo
mako-miyatake@g.ecc.u-tokyo.ac.jp

Yuji Sekiya
The University of Tokyo
sekiya@nc.u-tokyo.ac.jp

Koya Narumi
The University of Tokyo
narumi@akg.t.u-tokyo.ac.jp

Yoshihiro Kawahara
The University of Tokyo
kawahara@akg.t.u-tokyo.ac.jp

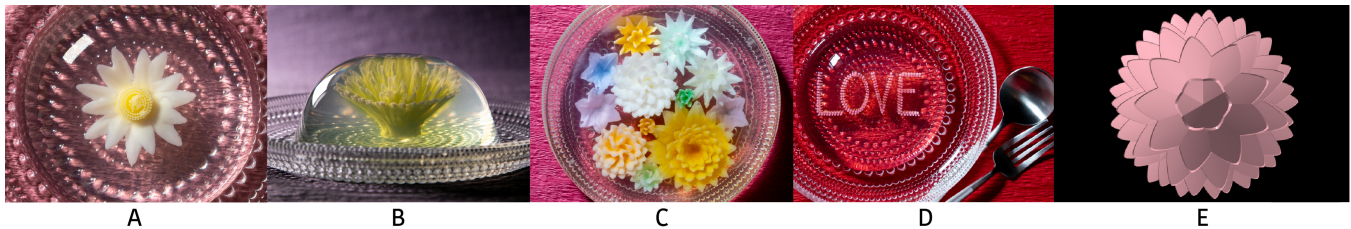


Figure 1: Flower jellies made with Flower Jelly Printer. (A) Marguerite, (B) Mum, (C) A cake with several types of flowers, (D) A printed message, and (E) A schematic preview of Dahlia in the computational design tool.

ABSTRACT

Flower jellies, a delicate dessert in which a flower-shaped jelly floats inside another clear jelly, fascinate people with both their beauty and elaborate construction. In efforts to simplify the challenging fabrication and enrich the design space of this dessert, we present Flower Jelly Printer: a printing device and design software for digitally fabricating flower jellies. Our design software lets users play with parameters and preview the resulting forms until achieving their desired shapes. We also developed slit injection printing that directly injects colored jelly into a base jelly, and shared several design examples to show the breadth of design possibilities. We hope to enable more people to design and create their own flower jellies while expanding access and the design space for digitally fabricated foods.

CCS CONCEPTS

• **Human-centered computing** → **User interface toolkits**; • **Applied computing** → *Computer-aided manufacturing*.

KEYWORDS

Digital food fabrication; food 3D printer; fabrication techniques; interactive food; food-interaction design.

ACM Reference Format:

Mako Miyatake, Koya Narumi, Yuji Sekiya, and Yoshihiro Kawahara. 2021. Demonstrating Flower Jelly Printer for Parametrically Designed Flower Jelly. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 8–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3411763.3451541>

1 INTRODUCTION

Flower jelly, as illustrated in Figure. 1, is an artistic dessert in which an intricate flower-shaped jelly floats inside a clear jelly matrix. Appearing just like real flowers suspended in glass, these handmade creations demonstrate patissiers' extremely skilled craftsmanship. Our motivation is to maximize the aesthetic possibilities and simplify the fabrication of this soft, fragile, and difficult to produce delicacy by means of digital fabrication techniques.

In the typical cooking process, patissiers commonly suffer from the following issues in the manual procedure: (1) Precise and accurate control of the hand while working with the soft material; (2) Prolonged concentration and effort required by the slow, manual process; and (3) Inability to undo or redo mistaken cuts and fills.

We believe that digital fabrication techniques are key to solving these problems. Although digital fabrication tools have been already introduced into many food fabrication processes, they cannot be applied to all types of food production. Unlike chocolates which solidify immediately at room temperature when extruded from a heated nozzle, hydrogel-based materials (e.g., gelatin and agar) take a long time to coagulate and retain the form, resulting in deformation while printing. Therefore, it is challenging to 3D print flower jellies using existing additive 3D food printing methods.

Instead, we present Flower Jelly Printer, a novel printing platform

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '21 Extended Abstracts, May 8–13, 2021, Yokohama, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3451541>

that not only addresses the challenges of manual fabrication procedures but also extends the design space of flower jellies. Our computational design tool allows users to parametrically play with the shape of the flower jelly and generates 3D previews until they find the exact parameters of their favorite designs. We overcome material limitations of existing food printers by developing a slit injection printing technique that directly injects colored jelly into a base jelly. Furthermore, we examined the material properties of jellies, introduced mechanical designs into the proposing printer, and built path planning algorithms to establish the digital fabrication of flower jellies.

2 RELATED WORK

Digital Food Fabrication

Recently, digital fabrication techniques have been introduced into food fabrication processes, represented by the fact that food 3D printers are increasingly available on market [1, 6]. This growing availability has promoted interest in research on digital food fabrication. For example, Digital Gastronomy leverages digital fabrication tools to create novel recipes such as 3D printed noodles and a CNC milled sweet potato [7].

Especially, extrusion-based additive manufacturing is frequently utilized in human-computer interaction (HCI) to achieve unique experiences with food. FoodFab [5] is the system that creates perceptual illusions with a food 3D printer by changing infill patterns and infill densities. Transformative Appetite [9] extrudes edible materials (e.g., gelatin, cellulose) to fabricate shape-changing edible films that transform into 3D shapes through water adsorption. Although these prior works have spread the design space of current digital food fabrication, their relatively simple extrusion processes are not suitable for the fabrication of flower jelly, which needs to build a delicate and tall structure with slowly curing and deformable materials like gelatin.

There are also several prior works on digital food fabrication that leverage subtractive manufacturing processes. This category includes CNC milling [7], stamping and grooving with digitally controlled jigs [8] or engraving with a laser cutter [2]. However, these subtractive fabrications are not very suitable for the soft and therefore fragile nature of jelly. This paper aims at developing a novel digital fabrication technique to widen the design freedom of current flower jellies by design iteration using CAD software and the precise and automated fabrication enabled by our printer.

3D Printing in the Granular Gel Support

Although they are not related to food, there are a few prior works on 3D printing materials in a gel support, which are similar to our approach. In the field of material science, Hinton et al. developed the technique to 3D print protein and polysaccharide hydrogels [3] for biological and medical applications. They used microparticle gel materials ground in a mixer or a centrifuge as a supporting bath to sustain the printing inks until they are fully cured. Likewise, Rapid Liquid Printing [4] utilized the granular gels as the supporting bath to 3D print large objects in any direction with faster speed.

In this paper, we also 3D print materials inside a gel support. The difference is that we inject colored jelly into the “bulk” (*i.e.*, not

granular) firm jelly, so we can take it out from the mold and eat the finished jelly right after it is printed while sustaining the pre-designed aesthetic shape.

3 FLOWER JELLY PRINTER

3.1 Overview

Flower Jelly Printer is a printing platform that allows users to design flower jellies and fabricate them precisely through digital design and fabrication. In this section, we will follow each step of our system: (a) Users design their desired flower jelly with our computational design tool; (b) Users fabricate custom tool attachments for the printer, export a G-code file, and prepare two jelly materials; (c) Our 3D printer prints the designed flower jelly by slit injection printing.

3.2 Computational Design Process

To support the trial-and-error design iteration, we built a parametric design interface with Rhinoceros and Grasshopper. As shown in Figure. 2, we provided a predefined library of example flower forms to suggest default parameters. After selecting their favorite type of flower from the library, users launch the software of the chosen type to customize it.

Next, users can design the shape of each petal (Figure. 2 left). Our software offers several predefined shapes with customizable design parameters to achieve a wide variety of shapes; from sharp, narrow petals like marguerites to rounded, broad ones like roses. Users can also customize the overall structure of the flower (Figure. 2 center). All floral components have the same axis of rotation of each flower and every group of petals are radial symmetrically arranged in each layer. Through the designing process, a 3D preview corresponding to the current set of parameters gives users a quick way to visualize the final result of their design. Once users are satisfied with the design, they can export an STL file for making the petal knife and a custom G-code file for the actual printing process of flower jellies.

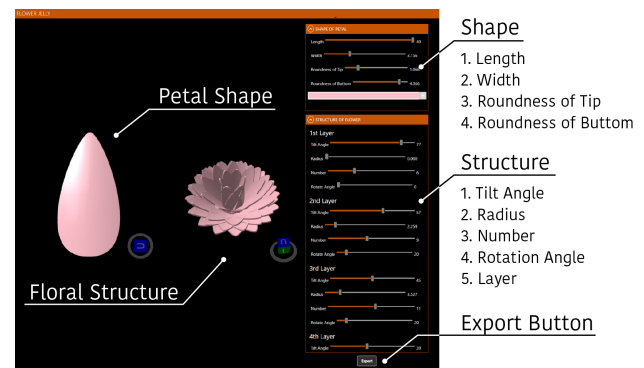


Figure 2: Design interface. The 3D preview of floral petal (left). The 3D preview of whole flower (center). Users can tune parameters for the shape of each petal (top right) and the whole appearance (bottom right), and iterate the design by checking the visualized model.

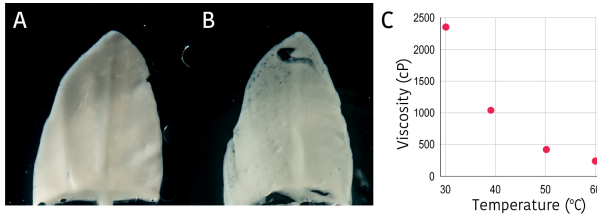


Figure 3: Viscosity of colored jelly. (A) No gaps in the final look of the petal at 30 °C. (B) A lack of uniformity at 60 °C. (C) The graph shows the relationship between viscosity and temperature of colored jelly.

3.3 Preparation Process

Before printing, users need to fabricate a custom petal knife and prepare jellies as printing materials.

The petal knife can be 3D printed from our design software. Once the design of the floral petal is finalized, our software generates an STL file of the petal knife that combines the designed petal shape, an injection needle, and a connection mechanism to a silicone tube. The petal knife works not only as a knife to make a slit but also as a needle to inject colored jelly into the base jelly.

To 3D print flower jelly, we prepared two different print materials: (1) base jelly, and (2) colored jelly. For the base jelly, we used 48 g *Pearl Agar*, 1 l flavored water, and 136 g sugar. For the colored jelly, we used 12 g *Jeleaf Jellyace* granular gelatin, 400 ml milk, 6 g agar, 100 g sugar, and food coloring.

Viscosity of Colored Jelly

Because the viscosity of the colored jelly has a large effect on the final look and finish of the petals, we experimentally determined the relationship between viscosity and gel temperature (shown in Figure. 3). Results clarified that 30 °C is the optimal temperature of the colored jelly. Figure. 3 (B) also shows how low viscosity in the colored jelly results in a lack of uniformity because the base jelly can push out the colored jelly, leaving gaps.

3.4 Printing Process

Slit Injection Printer

To implement the slit injection printing technique, we modified a 3D printer Ender 3 by adding a motorized turntable and a DIY

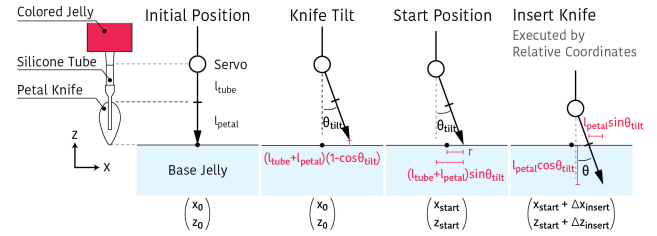


Figure 4: Printing motions. (A) Initial position. (B) Knife Tilt. (C) Start position. (D) Insert Knife.

paste extruder with a petal knife. This printer is compatible with the G-code toolpath compiled by our design software. We designed a custom paste extruder to change the angle of the tip during the slitting process (Figure. 5). We use a linear rail and a feed screw to convert the rotation of the motor (Nema 17 stepper motor) into the linear movement which pushes the plunger of the syringe (TELMO Syringe 30 ml). To enable changing the angle of the petal knife, we use a silicone tube (Leobro silicone tube 4 mm × 6 mm), which allows colored jelly to flow from the syringe to the petal knife regardless of knife angle. An MG90S servo motor mounted between the syringe and petal knife enables the knife to tilt to the desired angle. We also replaced the standard printer's heating bed with our DIY motorized turntable. We use a Nema 17 stepper motor controlled by a BigTreeTech SKR v1.4 mainboard to control the rotation of the turntable.

Path Planning Algorithm

Our software converts the parameters of the designed floral structure into the printing path, written as G-code. In order to make the finished flower jelly more aesthetically pleasing, printing paths are generated from the inside of the flower going outward. Before printing, users have to initialize the printer to set the initial position (x_0, z_0) .

1. *Knife Tilt* Tilt angles of the cutting knife are converted into the desired angle of the servo motor and then set using the M280 command of G-code via the mainboard.

2. *Start Position* The knife's start position (x_{start}, z_{start}) corresponds to the position of the bottom of the petal (Figure. 4 (C)). The coordinates are calculated by radius (r), tilt angle (θ_{tilt}), tube length (l_{tube}) and petal length (l_{petal}).

3. *Insert Knife* The path of knife insertion is determined by tilt

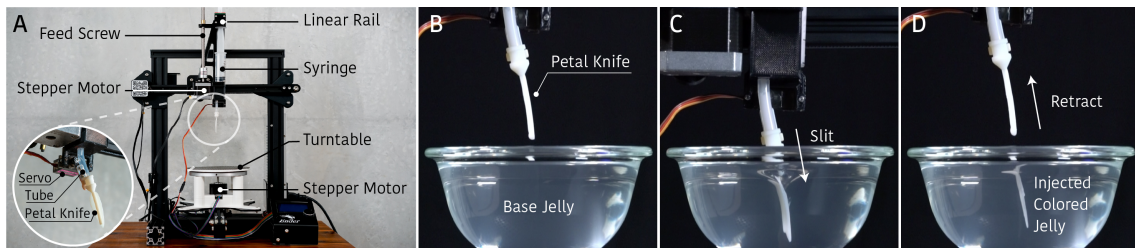


Figure 5: (A) The overview of the proposed printer. (B-D) The slit injection process. First, the petal knife makes a slit inside a base jelly. Then the printer retracts the knife while injecting a colored jelly.



Figure 6: Demonstrations of several printed patterns.

angle and petal length. This motion is operated by relative coordinates (Δx_{insert} , Δz_{insert}) from the coordinates of the start position (Figure. 4 (D)).

4. *Inject Colored Jelly & Pull Out Knife* While pulling the petal knife out from the base jelly, the printer pushes the syringe plunger to inject colored jelly. The length of extrusion (l_{length}) is converted from the volume of the petal (V_{petal}).

5. *Rotate Bowl* The rotate motion is executed (1) before, (2) during, and (3) after each layer printing process. (1) The motorized turntable rotates the desired angle to prevent petals in a specific layer from overlapping petals in another layer. (2) After each petal is printed by inserting motion, the turntable rotates the calculated angle ($\frac{360^\circ}{n}$) for the next petal because petals are arranged in radial symmetry. (3) To set the turntable to the initial position, the turntable rotates back the rotation angle ($-\theta_{\text{rotate}}$).

4 CONCLUSION

We presented the Flower Jelly Printer: a custom printing device and computational design tool for the design and digital fabrication of flower jellies. We implemented design software to allow users to play with design parameters and preview the resulting forms. Our newly developed slit injection printing technique directly injects colored jelly into a base jelly, which makes it possible to print of soft and fragile jelly structures. Our library of design examples demonstrates the aesthetic range of jellies that can be made with our system. We hope that more people will be able to enjoy creating and consuming these aesthetic desserts and that this work will open up new possibilities for personal design and fabrication of soft, gel-based foods.

ACKNOWLEDGMENTS

This work was supported by JST ERATO Grant Number JPM-JER1501, Japan. The authors would also like to thank Dr. Jie Qi for the support in the writing process; and Prof. Yasuaki Kakehi

and Prof. Takeo Igarashi for helpful comments.

REFERENCES

- [1] byFlow. 2015. *3D Food Printer | Focus | byFlow*. Retrieved September 17, 2020 from <https://www.3dbyflow.com/>
- [2] Kentaro Fukuchi, Kazuhiro Jo, Akifumi Tomiyama, and Shunsuke Takao. 2012. Laser Cooking: A Novel Culinary Technique for Dry Heating Using a Laser Cutter and Vision Technology. In *Proceedings of the ACM Multimedia 2012 Workshop on Multimedia for Cooking and Eating Activities* (Nara, Japan) (CEA '12). Association for Computing Machinery, New York, NY, USA, 55–58. <https://doi.org/10.1145/2390776.2390788>
- [3] Thomas J. Hinton, Quentin Jallerat, Rachelle N. Palchesko, Joon Hyung Park, Martin S. Grodzicki, Hao-Jan Shue, Mohamed H. Ramadan, Andrew R. Hudson, and Adam W. Feinberg. 2015. Three-dimensional printing of complex biological structures by freeform reversible embedding of suspended hydrogels. *Science Advances* 1, 9 (2015). <https://doi.org/10.1126/sciadv.1500758> arXiv:<https://advances.sciencemag.org/content/1/9/e1500758.full.pdf>
- [4] Self-Assembly Lab. 2017. *Rapid Liquid Printing – Self-Assembly Lab*. Retrieved September 17, 2020 from <https://selfassemblylab.mit.edu/rapid-liquid-printing>
- [5] Ying-Ju Lin, Parinya Punpongsonan, Xin Wen, Daisuke Iwai, Kosuke Sato, Marianna Obrist, and Stefanie Mueller. 2020. FoodFab: Creating Food Perception Illusions Using Food 3D Printing. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376421>
- [6] Natural Machines. 2015. *Natural Machines - Foodini*. Retrieved September 17, 2020 from <https://www.naturalmachines.com/foodini>
- [7] Moran Mizrahi, Amos Golan, Ariel Bezaleli Mizrahi, Rotem Gruber, Alexander Zoender Lachnise, and Amit Zoran. 2016. Digital Gastronomy: Methods & Recipes for Hybrid Cooking. In *Proceedings of the 29th Annual Symposium on User Interface Software and Technology* (Tokyo, Japan) (UIST '16). Association for Computing Machinery, New York, NY, USA, 541–552. <https://doi.org/10.1145/2984511.2984528>
- [8] Ye Tao, Youngwook Do, Humphrey Yang, Yi-Chin Lee, Guanyun Wang, Catherine Mondao, Jianxun Cui, Wen Wang, and Lining Yao. 2019. Morphlour: Personalized Flour-Based Morphing Food Induced by Dehydration or Hydration Method. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology* (New Orleans, LA, USA) (UIST '19). Association for Computing Machinery, New York, NY, USA, 329–340. <https://doi.org/10.1145/3332165.3347949>
- [9] Wen Wang, Lining Yao, Teng Zhang, Chin-Yi Cheng, Daniel Levine, and Hiroshi Ishii. 2017. Transformative Appetite: Shape-Changing Food Transforms from 2D to 3D by Water Interaction through Cooking. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 6123–6132. <https://doi.org/10.1145/3025453.3026019>